Year 11 Mathematics

Graphs 5

3.1 Linear Simultaneous Equations

Linear Simultaneous Equations

STARTER · 5 MIN

Date: _____ **Title:** Linear Simultaneous Equations

LEARNING OBJECTIVES

- Solve simultaneous equations by drawing graphs
- Solve simultaneous equations by elimination
- Solve simultaneous equations by substitution and in problems

STARTER *Answer in your book. No calculators.*

1. Find y when x = 2 for y = 3x − 1
2. Solve 2x = 14
3. Work out 3y − (−2y)
4. Is (2, 5) on the line y = 2x + 1?

Starter — Answers

STARTER

Tick or correct in green pen

1. $y = 6 - 1 = \mathbf{5}$

2. $x = \mathbf{7}$

3. $3y + 2y = \mathbf{5y}$

4. $2 \times 2 + 1 = 5$, **Yes**

Success Criteria

SUCCESS CRITERIA

By the end of this lesson:

BRONZE	I can solve simultaneous equations by drawing graphs	☐
SILVER	I can solve simultaneous equations by elimination	☐
GOLD	I can solve simultaneous equations by substitution and in problems	☐

Tick each box in the plenary when you can do it on your own.

Introduction — Simultaneous Equations

INTRODUCTION · 3 MIN

Two equations, two unknowns: find the values that work in both.

Graphs

Draw both lines. The solution is where they cross.

Cross at (2, 3): $x = 2$, $y = 3$

Elimination

Add or subtract to remove one letter.

Same sign: subtract. Different sign: add.

Substitution

Replace y with an expression in x.

$y = 2x$, $x + y = 9 \rightarrow 3x = 9$

Check

Substitute both values into both equations.

Both equations must work

Examples 1 and 2 — Solving with Graphs

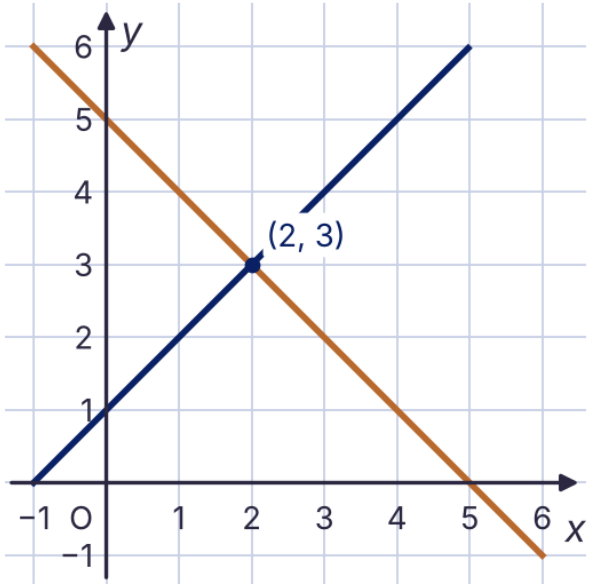
EXAMPLES · 5 MIN

BRONZE

Watch and copy into your book

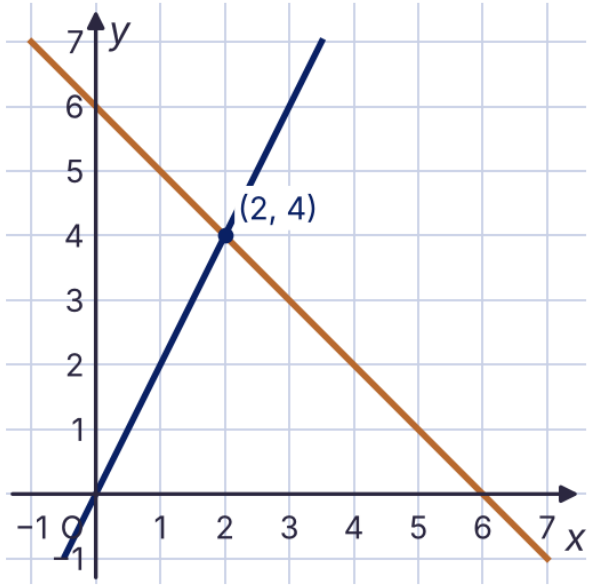
Example 1

Use the graph to solve $y = x + 1$ and $y = 5 - x$.



Example 2

Use the graph to solve $y = 2x$ and $x + y = 6$.



Examples 1 and 2 — Corrections

CORRECTIONS

BRONZE

Check your copy and correct in green pen

Example 1

The lines cross at (2, 3)

x = 2

y = 3

Tip: read the point where the lines cross.

Example 2

The lines cross at (2, 4)

x = 2

y = 4

Tip: check: 2 + 4 = 6 ✓

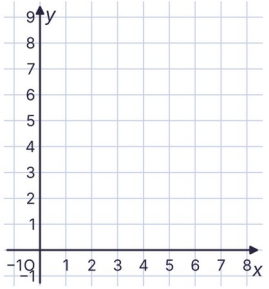
Task 1

PRACTICE · 9 MIN

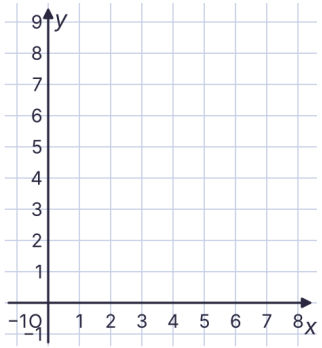
BRONZE

Draw both lines for x from 0 to 6, then solve

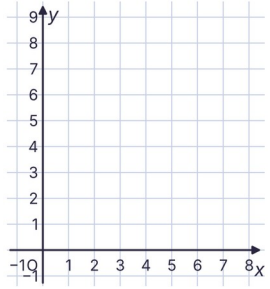
1 $y = x + 2$ and $y = 8 - x$



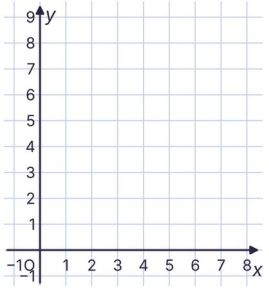
2 $y = 2x$ and $y = x + 3$



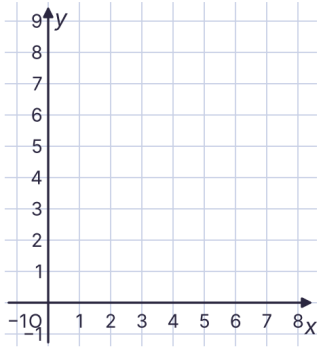
3 $y = 3x - 2$ and $y = x + 2$



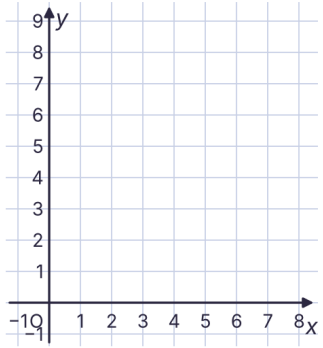
4 $x + y = 7$ and $y = x + 1$



5 $y = 2x - 1$ and $y = 5$



★ $y = 9 - 2x$ and $y = x$



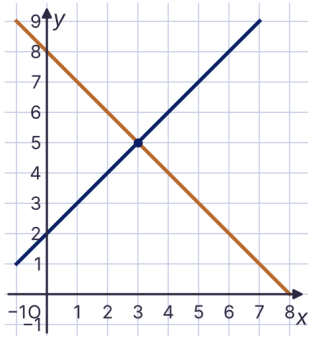
Task 1 — Answers

CORRECTIONS

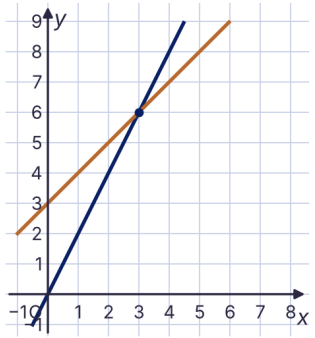
BRONZE

Tick or correct in green pen

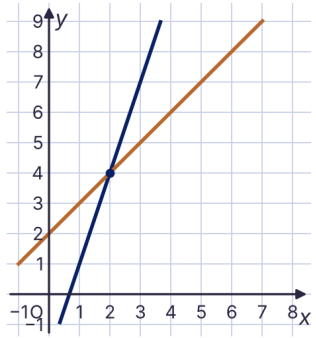
1 x = 3, y = 5



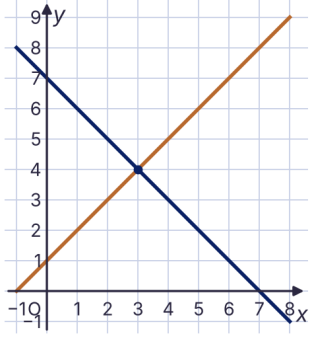
2 x = 3, y = 6



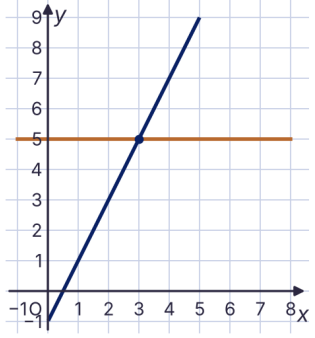
3 x = 2, y = 4



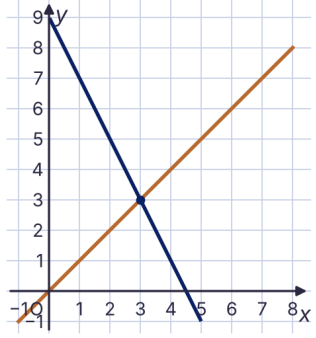
4 x = 3, y = 4



5 x = 3, y = 5



★ x = 3, y = 3



Examples 3 and 4 — Elimination

EXAMPLES · 5 MIN

SILVER

Watch and copy into your book

Hint same signs: subtract different signs: add

Example 3

Solve

$3x + y = 11$

$x + y = 5$

Example 4

Solve

$2x + 3y = 12$

$4x - 3y = 6$

Examples 3 and 4 — Corrections

CORRECTIONS

SILVER

Check your copy and correct in green pen

Example 3

Subtract: $2x = 6$

$x = 3$

$3 + y = 5$

$y = 2$

Tip: same y terms: subtract.

Example 4

Add: $6x = 18$

$x = 3$

$6 + 3y = 12$

$y = 2$

Tip: opposite signs: add.

Task 2

PRACTICE · 9 MIN

SILVER

Solve each pair of simultaneous equations

Hint same signs: subtract different signs: add

1. $x + y = 10, \quad x - y = 4$
2. $2x + y = 9, \quad x + y = 6$
3. $3x + 2y = 16, \quad x - 2y = 0$
4. $5x + y = 17, \quad 3x + y = 11$
5. $4x + 3y = 25, \quad 2x + 3y = 17$

Task 2 — Answers

CORRECTIONS

SILVER

Tick or correct in green pen

1. $2x = 14, \quad x = 7, \quad y = \mathbf{3}$

2. $x = 3, \quad y = \mathbf{3}$

3. $4x = 16, \quad x = 4, \quad y = \mathbf{2}$

4. $2x = 6, \quad x = 3, \quad y = \mathbf{2}$

5. $2x = 8, \quad x = 4, \quad y = \mathbf{3}$

Examples 5 and 6 — Harder Elimination and Problems

EXAMPLES · 5 MIN

GOLD

Exam-style: show all your working

Hint same signs: subtract different signs: add

Example 5

(3 marks)

Solve

$2x + 3y = 13$

$3x + 2y = 12$

Example 6

(4 marks)

2 pens and 3 rulers cost £3.90.

1 pen and 2 rulers cost £2.30.

Find the cost of each.

Examples 5 and 6 — Corrections

CORRECTIONS

GOLD

Check your copy and correct in green pen

Example 5

×3: $6x + 9y = 39$

×2: $6x + 4y = 24$

Subtract: $5y = 15$

$y = 3$

$2x + 9 = 13$

$x = 2$

Tip: multiply to make the x terms match.

Example 6

$2p + 3r = 390$

$p + 2r = 230$

×2: $2p + 4r = 460$

Subtract: $r = 70$

$p = 230 - 140$

Pen 90p and ruler 70p

Tip: write two equations first.

Task 3

PRACTICE · 9 MIN

GOLD

Exam-style: show all your working

Hint same signs: subtract different signs: add

1. Solve $3x + 2y = 19$ and $x + 4y = 13$ (3 marks)

2. Solve $5x - 2y = 11$ and $3x + 4y = 17$ (3 marks)

3. Solve $y = 2x + 1$ and $3x + y = 16$ (substitution) (3 marks)

4. 3 adult and 2 child tickets cost £36. 1 adult and 4 child tickets cost £32. Find the price of each. (4 marks)

5. The sum of two numbers is 25 and their difference is 7. Find the numbers. (3 marks)

Task 3 — Answers

CORRECTIONS

GOLD

Tick or correct in green pen

1. $\times 2: \; 6x + 4y = 38, \quad 5x = 25, \quad x = 5, \quad y = \mathbf{2}$

2. $\times 2: \; 10x - 4y = 22, \quad 13x = 39, \quad x = 3, \quad y = \mathbf{2}$

3. $3x + 2x + 1 = 16, \quad x = 3, \quad y = \mathbf{7}$

4. Adult = £8, Child = **£6**

5. $2x = 32, \quad x = 16, \quad y = \mathbf{9}$

Plenary — Exit Ticket

PLENARY · 5 MIN

Answer in your book. Try all three.

BRONZE Solve $x + y = 8$ and $x - y = 2$

SILVER Solve $2x + y = 7$ and $x + y = 4$

GOLD Solve $2x + 3y = 8$ and $3x + 2y = 7$

Reflect: Look back at the learning objectives. Which ones can you do now?

Plenary — Answers

PLENARY

Tick or correct in green pen

BRONZE $2x = 10, \quad x = 5, \quad y = \mathbf{3}$

SILVER $x = 3, \quad y = \mathbf{1}$

GOLD $x = 1, \quad y = \mathbf{2}$

Exam Practice — Graphs and Elimination

EXAM PRACTICE · 10 MIN

BRONZE

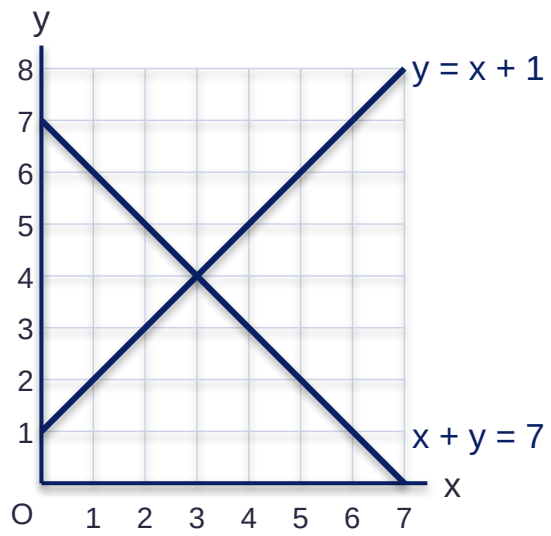
IGCSE exam-style: answer on the line, show all your working

Question 1

(2 marks)

Use the graph to solve

$y = x + 1$ and $x + y = 7$



x = ... y = ...

Question 2

(3 marks)

Solve the simultaneous equations

$x + y = 9$

$x - y = 3$

x = ... y = ...

Exam Practice — Elimination

EXAM PRACTICE · 10 MIN

SILVER

IGCSE exam-style: show all your working

Hint same signs: subtract different signs: add

Question 3

(3 marks)

Solve the simultaneous equations

$3x + 2y = 18$

$x + 2y = 10$

x = ... y = ...

Question 4

(3 marks)

Solve the simultaneous equations

$4x + y = 14$

$2x - y = 4$

x = ... y = ...

Exam Practice — Harder Elimination and Problems

EXAM PRACTICE · 10 MIN

GOLD

IGCSE exam-style: show all your working

Hint same signs: subtract different signs: add

Question 5

(4 marks)

Solve the simultaneous equations

$5x + 2y = 16$

$3x - 4y = 20$

x = ... y = ...

Question 6

(4 marks)

4 coffees and 3 teas cost £14.10

2 coffees and 5 teas cost £12.30

Work out the cost of one coffee and the cost of one tea.

..... coffee £..... tea £

Exam Practice — Mark Scheme

CORRECTIONS

BRONZE

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 1

lines cross at (3, 4) — M1

x = 3, y = 4 — A1

Tip: read the point where the lines cross.

Question 2

add: 2x = 12 — M1

x = 6 — A1

6 + y = 9, y = 3 — A1

Tip: check in the second equation: 6 − 3 = 3.

Exam Practice — Mark Scheme

CORRECTIONS

SILVER

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 3

subtract: 2x = 8	M1
x = 4	A1
4 + 2y = 10, y = 3	A1

Tip: same 2y in both: subtract.

Question 4

add: 6x = 18	M1
x = 3	A1
12 + y = 14, y = 2	A1

Tip: +y and −y: add.

Exam Practice — Mark Scheme

CORRECTIONS

GOLD

Mark your work: M = method mark, A = accuracy mark, B = independent mark

Question 5

× 2: 10x + 4y = 32	M1
add: 13x = 52	M1
x = 4	A1
20 + 2y = 16, y = −2	A1

Tip: make the y terms match, then add.

Question 6

4c + 3t = 14.10, 2c + 5t = 12.30	M1
× 2: 4c + 10t = 24.60	M1
subtract: 7t = 10.50, t = £1.50	A1
4c = 9.60, c = £2.40	A1

Tip: write two equations first.